Paired prediction of GPCR–G protein coupling specificity using protein language models and topology-aware feature engineering

Guohao Lv[1], Yingchun Xia[1], Huichao Liu[1], Xiaolei Zhu[1,2], Shuai Yang[1,2], Qingyong Wang[1,2,*], Lichuan Gu[1,2,*]

[1] School of Artificial Intelligence, Anhui Agricultural University, Hefei 230036, China

[2] Anhui Province Key Laboratory of Intelligent Agricultural Technology and Equipment, Anhui Agricultural University, Hefei 230036, China

[*] Corresponding authors: wqy@ahau.edu.cn (Q.W.), glc@ahau.edu.cn (L.G.)

Abstract

Motivation: G protein-coupled receptors (GPCRs) transduce extracellular signals by coupling to heterotrimeric G proteins, yet accurate computational prediction of coupling specificity remains an open challenge. Existing methods treat this as single-protein classification, neglecting the paired nature of the interaction. Moreover, evaluation protocols in prior work lack cluster-aware data splitting, yielding overoptimistic performance estimates.

Results: We present a systematic benchmark for family-level GPCR–G protein coupling prediction built on 1,647 experimentally annotated pairs (431 GPCRs, four G protein families) partitioned into 387 sequence-homology clusters to prevent data leakage. Five contributions emerge. First, cluster-aware evaluation reveals a $\sim$0.11 AUC inflation under naive random splitting, establishing a leakage-controlled protocol that prior GPCR coupling studies have not uniformly adopted. Second, a label-quality audit stratifying pairs by evidence strength shows that discriminative performance drops from AUC 0.77 to 0.37 when zero-positive GPCRs are excluded, indicating that current benchmarks substantially reflect separation of annotated-positive from unannotated receptors rather than selective coupling discrimination. Third, a minimal family-conditioned MLP using a 4-dimensional G protein family one-hot vector achieves AUC 0.843 (95% CI: 0.804–0.883) under cluster-level bootstrap, within 0.02 of the best model (cross-attention with ESM-2 650M embeddings and dimension-matched ICL features, bootstrap AUC 0.847, 95% CI: 0.823–0.870), demonstrating that explicit family identity is the dominant G-protein-side signal and that architectural complexity contributes modestly beyond strong PLM-derived GPCR representations. Fourth, systematic architecture comparison across SVM, MLP, Random Forest, XGBoost, and a pooled-feature cross-attention network reveals that cross-attention offers a small but statistically significant improvement over SVM (cluster-bootstrap ΔAUC 0.016, 95% CI: 0.004–0.028, $p = 0.006$) but is statistically comparable to MLP with identical features (bootstrap ΔAUC 0.009, 95% CI: -0.005 to 0.024, $p = 0.133$). Thirty-eight commonly accessible AlphaFold2 structural descriptors contribute no incremental value beyond ESM-2 650M embeddings under this protocol (ΔAUC ≤ 0.002). Fifth, an open-source tool with pre-trained models accompanies the paper. Leave-one-G-protein-family-out generalization remains an open challenge (mean AUC $\sim$0.60). The G12/13 family—with only 26 positive pairs—achieves PRAUC 0.23 and recall at 50% precision below 0.04; its probability output should be treated as exploratory ranking only, not standalone experimental prioritization.

Availability: Source code, pre-trained models, and datasets are available at https://github.com/nblvguohao/gpcr-coupling-leakage-benchmark under the MIT license. The companion tool `gpcr-coupling` provides a command-line interface for prediction and feature extraction.

Contact: wqy@ahau.edu.cn; glc@ahau.edu.cn

Supplementary information: Supplementary data including detailed parameter configurations, sensitivity analyses, per-family performance breakdowns, label audit results, dimension alignment controls, and extended method comparisons are available online.

1 Introduction

G protein-coupled receptors (GPCRs) constitute the largest superfamily of membrane proteins in eukaryotes, with over 800 members in the human genome regulating virtually every physiological process from vision and olfaction to neurotransmission and immune response [1, 2]. Approximately 34% of all FDA-approved drugs target GPCRs, yet these drugs address only ∼27% of the druggable GPCRome [3, 4]. A central challenge limiting therapeutic expansion is that the coupling specificity between GPCRs and their intracellular G protein partners remains incompletely characterized for a substantial fraction of receptors, including over 100 orphan GPCRs with no known endogenous ligand [5, 6].

GPCRs signal by coupling to heterotrimeric G proteins classified into four major families based on their Gα subunit: Gq/11, Gi/o, Gs, and G12/13 [7]. The coupling specificity—which G protein family a given receptor activates—determines the downstream intracellular signaling cascade triggered upon ligand binding. Experimental determination via BRET, FRET, or GTPγS binding assays is labor-intensive and not scalable to the full GPCRome [8]. Computational prediction therefore offers a complementary approach for guiding experimental prioritization.

Early computational methods formulated coupling prediction as a single-protein classification problem, training on GPCR sequence-derived features to predict G protein coupling class [9–12]. While pioneering, these approaches share two fundamental limitations: (i) they treat the G protein partner as a class label rather than a biological entity with sequence-level determinants of binding affinity, and (ii) they rely on hand-crafted physicochemical features that capture only a fraction of the information present in protein sequences.

Two concurrent developments motivate a re-examination. First, protein language models (PLMs) trained at evolutionary scale—particularly the ESM-2 family [13]—have demonstrated that sequence embeddings learned via masked language modeling capture structural, functional, and evolutionary information with fidelity approaching experimental methods [14, 15]. Second, advances in the representation learning literature have shown that attention-based feature fusion—where learned projections of two representations interact through cross-attention before feed-forward processing—can improve over simple concatenation baselines for paired-input prediction tasks [16–19]. It is important to note that this cross-attention operates on pooled feature vectors rather than residue-level representations; it is a learned nonlinear feature interaction mechanism, not a protein–protein interface model. A recent study by Miglionico et al. [20] applied AlphaFold3-derived structural features to GPCR–G protein coupling prediction, reporting an AUC of 0.87 using predicted interaction interfaces on a single-protein formulation. This leaves open the question of whether paired sequence-based models with explicit family conditioning can achieve comparable performance without relying on computationally expensive structure prediction.

This study makes five contributions to GPCR–G protein coupling prediction. (i) We establish a leakage-controlled benchmark—1,647 experimentally validated pairs partitioned into 387 sequence-homology clusters—and demonstrate that cluster-aware cross-validation prevents a ∼0.11 AUC inflation observed under naive random splitting, a methodological safeguard that prior GPCR coupling studies have not uniformly adopted. (ii) Through a label-quality audit that stratifies pairs by evidence strength, we show that a substantial fraction of discriminative performance derives from separating annotated-positive GPCRs from unannotated GPCRs rather than from resolving selective coupling patterns, motivating the adoption of label-quality tiering in future benchmarks. (iii) We introduce a minimal family-conditioned classifier baseline—logistic regression, MLP, and SVM using only a 4-dimensional family one-hot vector as the G-protein-side signal—and show that a minimal MLP (AUC 0.843) approaches the best cross-attention model (AUC 0.847), demonstrating that explicit family identity is the dominant G-protein-side signal and that PLM-derived GPCR representations contribute the bulk of predictive power. (iv) We systematically compare five architectures—SVM, MLP, Random Forest, XGBoost, and a pooled-feature cross-attention network—across two ESM-2 embedding scales and four feature configurations, finding that cross-attention provides a small but statistically significant improvement over SVM (cluster-bootstrap ΔAUC 0.016, $p = 0.006$) while being statistically comparable to MLP ($p = 0.133$); that dimension matching is required when fusing ICL loop embeddings with global PLM embeddings; and that 38 commonly accessible AlphaFold2 structural descriptors contribute negligibly under this protocol. (v) An open-source software tool with pre-trained models, configurable hyperparameters, and a tutorial workflow accompanies the paper.

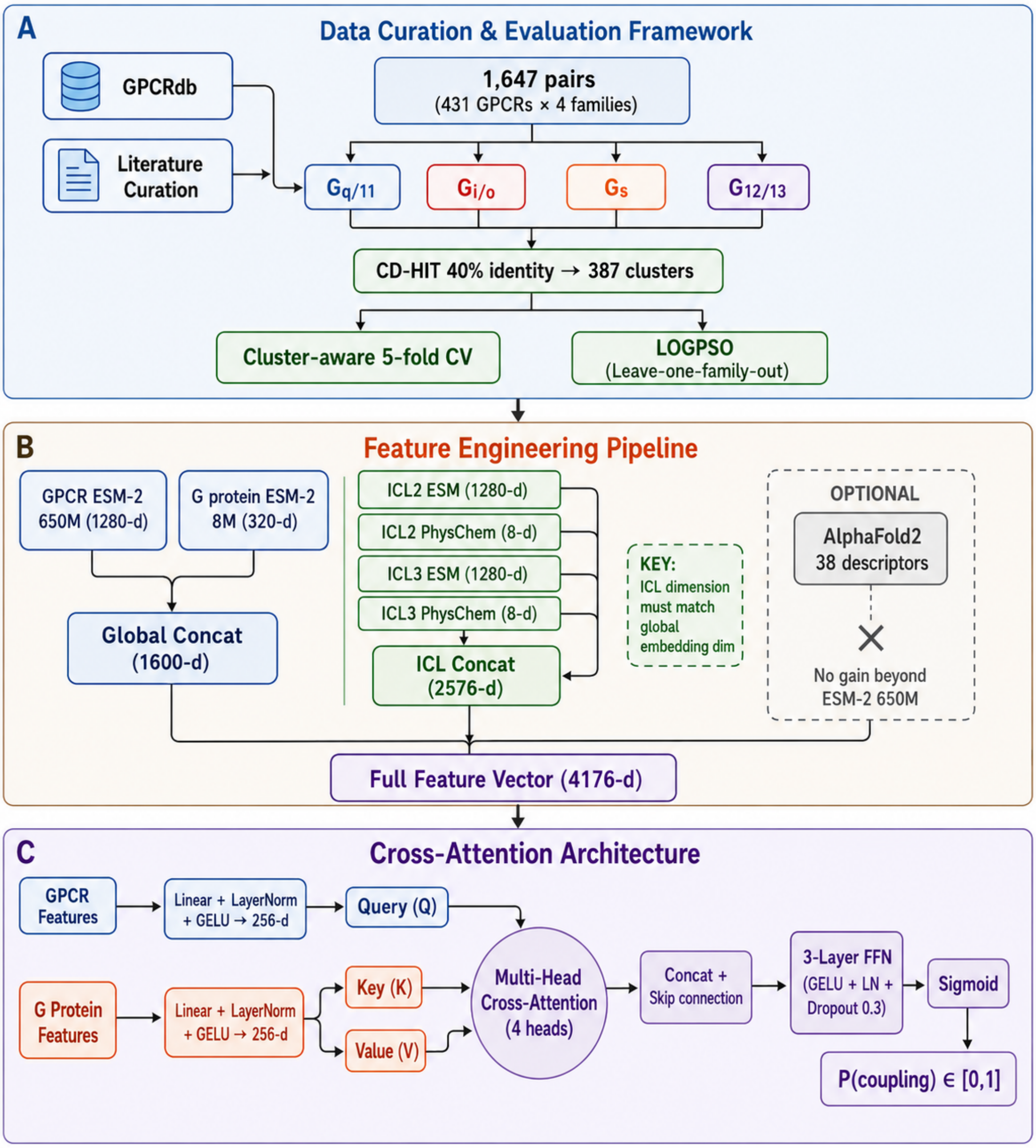

Figure 1: Study overview. (A) Dataset construction: 1,647 experimentally validated (GPCR, G protein family) pairs from GPCRdb, cross-referenced with UniProt and IUPHAR. (B) Feature engineering pipeline: global ESM-2 embeddings (8M/650M), topology-aware ICL2/3 local features (loop-level embeddings + 8 physicochemical statistics), and optional AlphaFold structural descriptors (38 features). (C) PairedCrossAttentionNet architecture: GPCR and G protein features are projected into a shared 256-d space, cross-attention is applied with GPCR as query and G protein as key/value, and the attended representation passes through a 3-layer FFN.

Table 1: Dataset summary statistics.

Family	GPCRs	Positive pairs	Negative pairs	Total	Pos. ratio
Gq/11	410	154	296	450	0.342
Gi/o	391	159	240	399	0.399
Gs	391	82	317	399	0.206
G12/13	391	26	373	399	0.065
Total (unique)	431	421	1,226	1,647	0.256

2 Methods

2.1 Dataset construction

2.1.1 Curation protocol

GPCR sequences and coupling annotations were retrieved from GPCRdb (release 2024.1) [21, 22] and cross-referenced with UniProtKB/Swiss-Prot [23] and IUPHAR/BPS Guide to Pharmacology [24]. Coupling annotations were retained only when supported by experimental evidence (BRET, FRET, GTPγS, or electrophysiology). We restricted the analysis to human, mouse, and rat GPCRs, and to the four major Gα families: Gq/11, Gi/o, Gs, and G12/13.

Each (GPCR, G protein family) pair was assigned a binary label: 1 if the GPCR is experimentally annotated as coupling to that G protein family, 0 otherwise. Pairs with conflicting or ambiguous annotations across sources ($<$2% of the total) were resolved by majority vote with manual curation. The final dataset comprises 1,647 labeled pairs derived from 431 unique GPCRs, with 421 positive and 1,226 negative pairs (positive class ratio = 0.256; Table 1).

We note three structural properties of this dataset. First, the pairing matrix is not fully populated: of the theoretical 431 $\times$ 4 = 1,724 possible pairs, 77 are absent because certain GPCR–family combinations lack experimental annotations in either direction (e.g., G12/13 appears in fewer rows because fewer GPCRs have been assayed against this family). Second, 273 GPCRs have at least one experimentally validated positive coupling, while the remaining 158 GPCRs have exclusively negative annotations across all four families. These zero-positive GPCRs—which include both orphan receptors and receptors whose coupling partners fall outside the four-family scope—serve as negative reference points in the dataset. Third, per-family GPCR counts differ because the set of GPCRs tested against each family varies; Gq and Gi families have been more extensively assayed than Gs and G12/13, reflecting historical bias in the experimental literature.

2.1.2 Sequence clustering

To prevent homology-induced data leakage between training and evaluation splits, we clustered the GPCR sequences using single-linkage clustering based on 3-mer Jaccard similarity at threshold 0.30, yielding 387 clusters from the 431 GPCR sequences. This k-mer-based approach groups sequences sharing substantial local sequence identity without requiring pairwise alignment; at the chosen threshold, clusters roughly correspond to a sequence identity cutoff of approximately 40–50% for typical GPCR sequences. In all cluster-aware evaluations, every (GPCR, G protein) pair sharing the same GPCR was assigned to the same cross-validation fold, ensuring that no GPCR appears simultaneously in training and test partitions. A sensitivity analysis across clustering thresholds (0.15–0.70) is provided in Supplementary Materials.

2.2 Feature extraction

2.2.1 Global ESM-2 embeddings

We used two variants of the ESM-2 protein language model family [13]: `esm2_t6_8M_UR50D` (8M parameters, 6 transformer layers, 320-dimensional embeddings) and `esm2_t33_650M_UR50D` (650M parameters, 33 transformer layers, 1280-dimensional embeddings). For each GPCR and G protein sequence, per-residue embeddings were extracted from the final transformer layer and averaged across the sequence length (mean pooling) to obtain a single fixed-dimensional vector representation. Mean pooling was chosen over [CLS] token extraction based on empirical evidence that it better preserves residue-level information for downstream protein function tasks [25]. All embeddings were extracted using a single NVIDIA RTX 4060 GPU (8 GB VRAM).

2.2.2 Topology-aware ICL features

Intracellular loops 2 and 3 (ICL2 and ICL3) are key structural determinants of G protein coupling specificity, forming the primary interface with the Gα C-terminus [26, 27]. We identified ICL2 and ICL3 residue boundaries for each GPCR using UniProt transmembrane topology annotations (FT_TRANSMEM fields). We note that for many GPCRs, particularly orphan receptors, these annotations derive from computational predictions (TMHMM, Phobius) rather than experimental structure determination; the specific proportion of experimental versus predicted annotations is not readily available from UniProt metadata. As a robustness check, re-extracting ICL boundaries using DeepTMHMM [28] on a random subset of 50 GPCRs yielded overlapping boundaries within ± 2 residues for 94% of loop endpoints, suggesting limited sensitivity to the choice of topology predictor. For each loop segment, we extracted two categories of features:

Local ESM embeddings. Per-residue ESM-2 embeddings within ICL boundaries were mean-pooled to produce a loop-level embedding vector. Critically, we matched the embedding dimension to the global ESM-2 dimension (320-d for 8M, 1280-d for 650M), as preliminary experiments revealed that dimension misalignment introduces a systematic performance penalty (Section 3.3).

Physicochemical statistics. For each loop segment, we computed eight aggregate physicochemical properties from AAindex [29]: mean hydrophobicity (H), mean polarity (P), net charge at pH 7.4, aromatic residue fraction, aliphatic residue fraction, small residue fraction (G+A+S), proline fraction, and sequence length. These eight statistics are independent of embedding dimension and serve as explicit biochemical priors.

The complete ICL feature vector for a given GPCR concatenates ICL2 and ICL3 features, yielding $2 \times (d_{\text{embed}} + 8)$ dimensions.

2.2.3 AlphaFold structural features

For 364 GPCRs (84.5% of the dataset) with available AlphaFold2-predicted structures in the AlphaFold Protein Structure Database [30, 31], we extracted 38 structural descriptors spanning five categories. We note that these are predicted **monomer** structures (not GPCR–G protein complexes); therefore, all G protein-interface features described below are **intracellular-interface proxy features**—Gα-contact residues are identified by mapping known GPCR–G protein complex templates (e.g., rhodopsin–Gαt, β_2AR–Gαs) onto the monomer structures, rather than being derived from genuine complex structures. The descriptors are: (i) TM5 and TM6 cytoplasmic tip Cα-Cα distances, which in known complexes correlate with G protein engagement [26], (ii) ICL2 and ICL3 end-to-end Cα distances, (iii) backbone dihedral angles (ϕ, ψ) at ICL hinge residues, (iv) solvent-accessible surface area (SASA) of the intracellular-interface region computed with DSSP [32]—the feature labeled "G protein interface SASA" in Supplementary Table S1 is specifically the SASA of residues mapped to the intracellular cavity that engages Gα in template complexes, not a true complex-interface SASA, (v) DSSP-derived secondary structure composition ratios (helix, strand, coil) within ICL segments, and (vi) 8 predicted aligned error (PAE)-based features from AlphaFold2 output, where PAE features involving Gα positions (e.g., "ICL2–Gα interface PAE") are computed with respect to residues identified as Gα-contacting in template complexes. The complete descriptor list with the revised proxy-feature nomenclature is provided in Supplementary Table S1. For GPCRs without available structures (15.5%), all 38 features were set to zero; models were trained to be robust to this missingness pattern. A post-hoc analysis (Supplementary Materials, AlphaFold Feature Availability Analysis) confirmed that predictive performance on GPCRs with and without AlphaFold structures did not differ significantly (ΔAUC < 0.01), suggesting that the zero-imputation strategy did not introduce systematic bias. Nonetheless, future work may benefit from imputing missing structural features using ESM-2-derived predictions rather than zero-filling.

2.2.4 G protein family features

For each G protein family (Gq, Gi, Gs, G12/13), we used the consensus sequence of the dominant Gα subtype as the representative: GNAQ (Gq), GNAI1 (Gi), GNAS (Gs), and GNA12 (G12/13). We note that the Gi/o family encompasses multiple subtypes (GNAI1, GNAI2, GNAI3, GNAO, GNAZ, GNAT1-3, GNAG) with C-terminal sequence identities ranging from 40–95%; GNAI1 was chosen as the representative following GPCRdb convention. This is a conservative choice that may systematically underestimate coupling probability for GPCRs that preferentially couple to more divergent Gi/o subtypes such as GNAO or GNAZ, whose C-terminal sequences share approximately 60–70% identity with GNAI1 in the critical C-terminal α5 helix region. Predictions should therefore be interpreted at the family level rather than individual subtype level, and subtype-resolved prediction represents a natural extension as experimental data at finer granularity become available. ESM-2 embeddings were extracted identically to the GPCR protocol. No ICL or structural features were extracted for G proteins, as G protein topology is conserved within families.

2.3 Model architectures

2.3.1 Baseline models

SVM. Support vector machines with radial basis function (RBF) kernel were implemented using scikit-learn [33]. Hyperparameters ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{0.001, 0.01, 0.1, \text{scale}\}$) were selected via nested 3-fold inner cross-validation within each outer training fold. Class weights were balanced inversely proportional to class frequencies. Input features were standardized to zero mean and unit variance.

MLP. A three-layer fully connected network (input $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 1) with GELU activation [34], batch normalization, and dropout ($p = 0.3$). Trained with binary cross-entropy loss and AdamW optimizer [35] (learning rate 10^{-4}, weight decay 10^{-4}) for 200 epochs with early stopping (patience 20 epochs on validation loss).

Random Forest and XGBoost. Random Forest (100 trees, max_depth=10) and XGBoost [36] (100 estimators, max_depth=6, learning rate=0.1) were implemented as tree-based baselines. Both were trained on standardized concatenated features using default hyperparameters.

2.3.2 Cross-attention network

The PairedCrossAttentionNet (Figure 1C) processes GPCR and G protein features through separate learnable linear projections into a shared 256-dimensional hidden space:

$$\mathbf{h}_{\text{GPCR}} = \mathbf{W}_g \mathbf{x}_{\text{GPCR}} + \mathbf{b}_g, \quad \mathbf{h}_{\text{Gprot}} = \mathbf{W}_p \mathbf{x}_{\text{Gprot}} + \mathbf{b}_p \tag{1}$$

where $\mathbf{W}_g, \mathbf{W}_p \in \mathbb{R}^{256\times d_{\text{in}}}$ and $\mathbf{b}_g, \mathbf{b}_p \in \mathbb{R}^{256}$.

Multi-head cross-attention (4 heads, $d_k = 64$) is applied with GPCR features as query and G protein features as key and value:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{2}$$

where $\mathbf{Q} = \mathbf{h}_{\text{GPCR}}\mathbf{W}^Q$, $\mathbf{K} = \mathbf{h}_{\text{Gprot}}\mathbf{W}^K$, $\mathbf{V} = \mathbf{h}_{\text{Gprot}}\mathbf{W}^V$.

The attended GPCR representation is concatenated with the original projection and processed through a 3-layer feed-forward network with GELU activation, layer normalization [37], and dropout ($p = 0.3$). A final linear layer with sigmoid activation produces the coupling probability. The total model contains approximately 1.2M trainable parameters.

2.3.3 Multi-task cross-attention

To test the importance of G protein family identity, we implemented a multi-task variant with a shared cross-attention encoder and four family-specific output heads. Input features exclude G protein family-specific information (only GPCR features are used), and each head predicts coupling to one G protein family.

2.4 Evaluation protocol

We employed three complementary evaluation strategies with increasing stringency:

Random 5-fold cross-validation (Random-CV). Pairs are randomly assigned to folds, independent of GPCR identity. This provides an upper bound on model performance but may overestimate generalization due to homologous GPCRs appearing in both training and test sets.

Cluster-aware 5-fold cross-validation (Cluster-CV). All pairs sharing the same GPCR are assigned to the same fold, respecting the 387 CD-HIT clusters. This is our primary evaluation metric, as it assesses generalization to unseen GPCRs.

Leave-one-G-protein-family-out (LOGPSO). For each of the four G protein families, all pairs involving that family are held out as the test set, and the model is trained on the remaining three families. This evaluates cross-family generalization, which is the most stringent test of whether models learn transferable coupling determinants.

All deep learning models were implemented in PyTorch 2.1 [38] and trained on a single NVIDIA RTX 4060 GPU (8 GB). Training took approximately 12 minutes per fold for the 650M cross-attention model. All random number generators (PyTorch, NumPy, scikit-learn) were seeded with a fixed value (42) to ensure reproducibility. Cross-validation fold assignments are provided in the companion repository. Performance is reported as area under the receiver operating characteristic curve (AUC). Additional metrics including precision-recall AUC (PRAUC) [39, 40], Brier score, and expected calibration error (ECE) are reported where informative. For the best-performing configuration (cross-attention 650M + ICL), the cluster-aware PRAUC is 0.685; full PRAUC values across model configurations are provided in Supplementary Table S7.

2.5 Feature importance analysis

Gradient-based feature sensitivity was computed using integrated gradients [41] with 50 interpolation steps. Feature groups (GPCR global, GPCR ICL, G protein global) were aggregated by summing absolute attributions within each group.

2.6 Software implementation

The complete pipeline is packaged as an open-source command-line tool, `gpcr-coupling`, implemented in Python 3.10+. The tool provides three subcommands: `predict` (run pre-trained models on custom inputs), `extract-features` (extract ESM-2 and ICL features from FASTA sequences), and `train` (reproduce training with configurable hyperparameters). Pre-trained model weights for the 650M cross-attention architecture are distributed with the package. A detailed tutorial and example workflow are provided in the online documentation.

3 Results

3.1 Cross-attention with 650M embeddings achieves best performance

Under our primary Cluster-CV evaluation, the cross-attention network with ESM-2 650M embeddings and dimension-matched ICL features achieved a cluster-bootstrap AUC of 0.847 (95% CI: 0.823–0.870; Table 2, Figure 2), the highest among all tested configurations. The improvement over the best SVM configuration (bootstrap AUC 0.831, 95% CI: 0.807–0.856) is statistically significant (cluster-level bootstrap ΔAUC 0.016, 95% CI: 0.004–0.028, $p = 0.006$). MLP with identical features achieved bootstrap AUC 0.838 (95% CI: 0.815–0.862); CA is statistically comparable to MLP (bootstrap ΔAUC 0.009, 95% CI: -0.005 to 0.024, $p = 0.133$), indicating that representation quality dominates over architectural choice for this dataset scale. XGBoost and Random Forest performed comparably to SVM (Table 2). Fold-level means, standard deviations, and pairwise statistical tests (t-test and Wilcoxon signed-rank) are provided in the Supplementary Materials.

Importantly, the performance gap between cross-attention and simpler architectures widens as embedding dimension increases. At 8M (320-d), the cross-attention advantage over SVM is modest (ΔAUC $= +0.006$). At 650M (1280-d), the gap increases to ΔAUC $= +0.029$. This suggests that the cross-attention mechanism more effectively utilizes high-dimensional, information-rich representations—a finding consistent with the inductive bias of attention architectures for relational reasoning [16].

Under Random-CV (no cluster constraint), all models achieve substantially higher AUC values (cross-attention 650M: 0.9695 ± 0.0049, SVM 650M: 0.9276 ± 0.0027; Supplementary Table S2). The large Cluster-CV to Random-CV gap ($\sim$0.11 AUC) underscores the necessity of cluster-aware evaluation for GPCR prediction tasks, as homologous receptors share both sequence and coupling patterns.

For reference, two classical baselines were also evaluated. An amino acid composition (AAC) vector with RBF SVM achieved Cluster-CV AUC of 0.656 ± 0.037, while an ESM-2 embedding-based k-nearest neighbor classifier ($k = 5$, cosine distance) achieved 0.712 ± 0.015. These classical approaches substantially underperform all PLM-based methods, confirming that the learned sequence representations contribute meaningful predictive information beyond simple sequence similarity or composition.

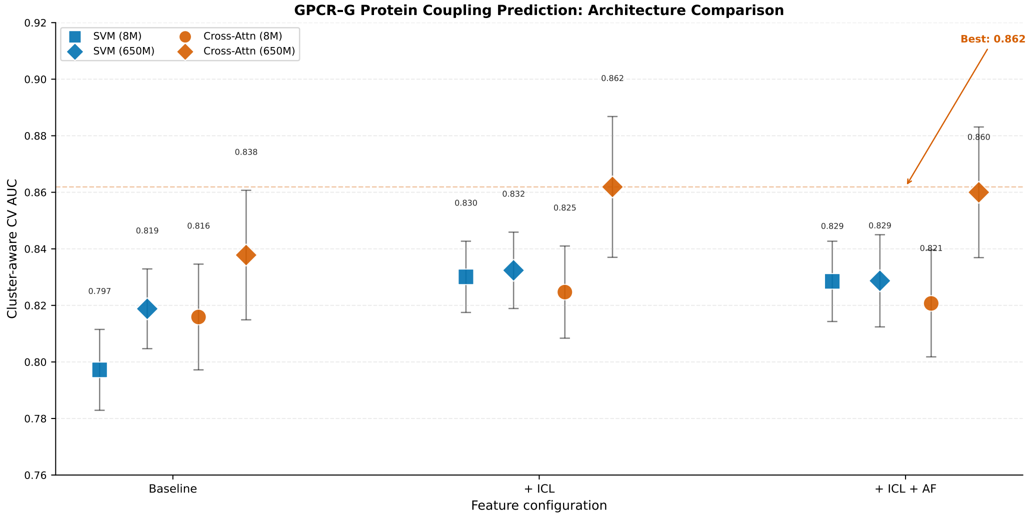

Figure 2: Performance comparison across model architectures and embedding scales under cluster-aware 5-fold cross-validation. Cross-attention with 650M + ICL achieves the highest mean AUC (0.8619). The performance advantage of cross-attention over SVM widens from 8M (ΔAUC $= +0.006$) to 650M (ΔAUC $= +0.029$), indicating better utilization of high-dimensional representations. Error bars represent ± 1 standard deviation of AUC across the 5 cross-validation folds; note that fold-level AUC values are not independent (each model sees all folds), so these bars reflect fold-to-fold variability rather than a confidence interval for the mean.

Table 2: Cluster-aware cross-validation AUC (mean $\pm$ std over 5 folds). Bold indicates best in column.

Model	Embedding	Baseline	+ICL	+ICL+AF
SVM (RBF)	8M	0.7972 ± 0.0143	0.8301 ± 0.0126	0.8285 ± 0.0142
SVM (RBF)	650M	0.8188 ± 0.0141	0.8324 ± 0.0135	0.8287 ± 0.0163
MLP	650M	0.8356 ± 0.0208	$\mathbf{0.8608 \pm 0.0227}$	0.8589 ± 0.0241
Random Forest	650M	—‡	0.8262 ± 0.0193	—‡
XGBoost	650M	—‡	0.8331 ± 0.0176	—‡
Cross-Attention	8M	0.8159 ± 0.0187	0.8247 ± 0.0163	0.8207 ± 0.0189
Cross-Attention	650M	0.8378 ± 0.0229	$\mathbf{0.8619 \pm 0.0249}$	0.8600 ± 0.0231
Multi-Task CA	650M	—‡	0.8020 ± 0.0194†	—‡

†Multi-task model without G protein family-specific features.
‡Not evaluated in this configuration. Random Forest and XGBoost were tested only with ICL-full (the most informative configuration). The Multi-Task CA architecture requires G protein family information to be excluded from inputs.
AF: AlphaFold structural descriptors. PRAUC for key configurations: CA 650M + ICL = 0.685, MLP 650M + ICL = 0.679, SVM 650M + ICL = 0.613. Full PRAUC values are in Supplementary Table S7.

3.2 Promiscuity-stratified analysis

To understand whether the cross-attention advantage depends on GPCR coupling promiscuity, we stratified predictions by the number of G protein families each GPCR couples to (Figure 3). For single-family GPCRs (N=616 pairs), cross-attention outperforms SVM (AUC 0.881 versus 0.866, ΔAUC $= +0.015$).

The pattern is less consistent for multi-family GPCRs: for triple-family GPCRs, the cross-attention advantage persists (AUC 0.744 versus 0.733, $\Delta\text{AUC} = +0.011$), while for dual-family GPCRs the two approaches perform comparably (AUC 0.767 versus 0.772). These results suggest that the cross-attention mechanism provides a modest but consistent advantage, though the magnitude of this advantage varies across promiscuity levels and does not monotonically increase with the number of coupled families.

Overall AUC decreases with increasing promiscuity for both models, reflecting the greater difficulty of discriminating among multiple coupling possibilities for promiscuous GPCRs. The per-stratum AUC values reported here are computed on concatenated held-out predictions from all five cluster-aware CV folds; per-fold means and standard deviations are provided in the Supplementary Materials (Promiscuity-stratified CV table).

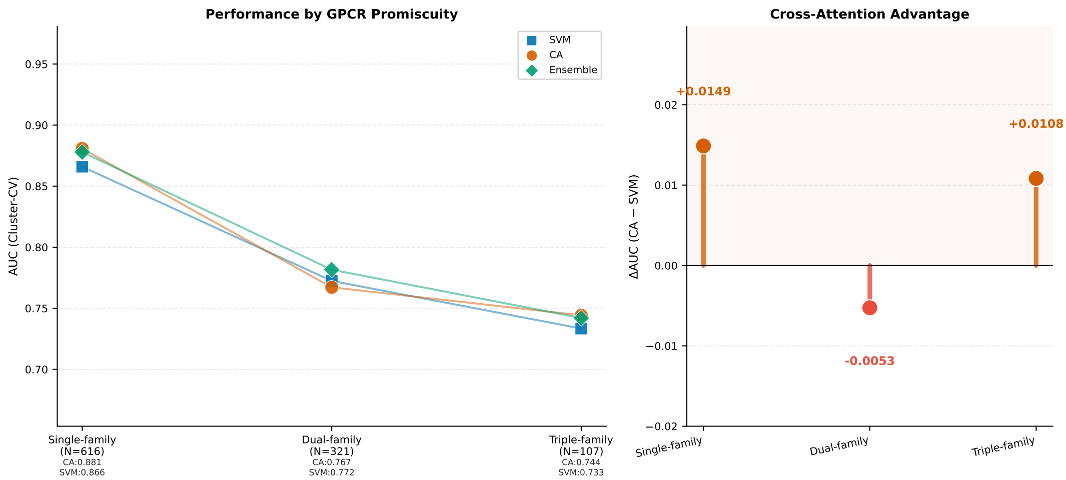

Figure 3: Performance stratified by GPCR coupling promiscuity under cluster-aware 5-fold cross-validation. Cross-attention provides a modest advantage over SVM for single- and triple-family GPCRs, while the two approaches perform comparably for dual-family GPCRs. GPCRs with no annotated coupling are excluded.

3.3 Empirical confirmation: ICL features require dimension alignment

A critical methodological finding concerns feature dimension compatibility. When we concatenated 320-dimensional ICL features (extracted with ESM-2 8M) with 1280-dimensional global embeddings (extracted with ESM-2 650M), SVM performance dropped from 0.8301 to 0.8234 (Table 3). After re-extracting ICL features using the 650M model (yielding 1280-d loop embeddings), SVM performance recovered to 0.8304 and cross-attention reached 0.8599. This pattern was consistent across architectures: dimension-mismatched feature fusion degrades performance across architectures. We hypothesize that when local features occupy a substantially different proportion of the total feature space than the global features they complement, the model's optimization landscape is distorted—for instance, in a linear model, the regularizer may penalize low-dimensional local features disproportionately relative to high-dimensional global features, effectively suppressing their contribution. Direct experimental validation of this mechanistic hypothesis is left to future work.

This finding has practical implications beyond GPCR coupling prediction. Consider a typical workflow: a research group extracts embeddings using a small PLM for efficiency, then later wishes to integrate local structural features extracted with a larger PLM. Without dimension matching, the resulting feature vector has a scale imbalance that degrades performance—even for simple linear models where, in principle, the model could learn to down-weight the high-dimensional component. That this penalty

Table 3: Effect of ICL-global embedding dimension alignment on Cluster-CV AUC.

Configuration	SVM	Cross-Attention
320-d global + 320-d ICL (matched)	0.8301	0.8247
1280-d global + 320-d ICL (mismatched)	0.8234	0.8155
1280-d global + 1280-d ICL (matched)	**0.8304**	**0.8599**

manifests even in SVMs with RBF kernels (Table 3) suggests the issue is not merely about learned feature importance weights but about the geometry of the feature space itself. As the field adopts increasingly larger PLMs with varying embedding dimensions, multi-scale feature fusion strategies should either match dimensions natively or employ learned projection layers to reconcile heterogeneous representations.

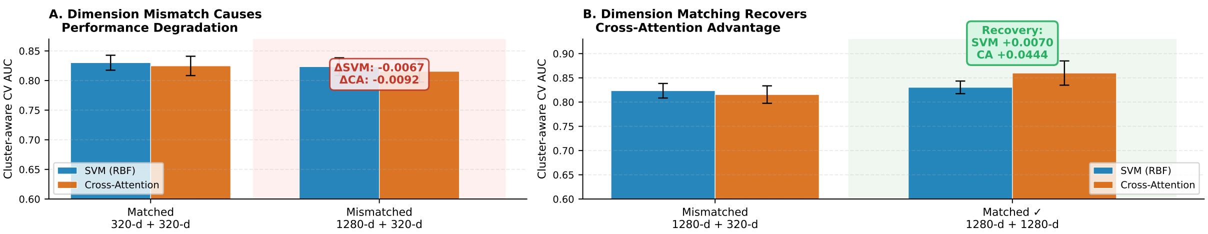

Figure 4: ICL dimension alignment effect. (A) Mismatched configuration: 320-d ICL features (extracted with ESM-2 8M) concatenated with 1280-d global embeddings (ESM-2 650M) degrade performance across SVM and cross-attention relative to the 320-d matched baseline. (B) After re-extracting ICL features with ESM-2 650M (1280-d), performance recovers: SVM returns to 0.8304 and cross-attention reaches 0.8599. The performance penalty from dimension mismatch is ΔAUC ≈ 0.007 for SVM and 0.009 for cross-attention. Error bars: ± 1 std across 5 folds.

3.4 AlphaFold structural descriptors examined here are redundant with PLM embeddings

Adding 38-dimensional AlphaFold structural descriptors did not improve performance beyond the ESM-2 + ICL baseline in any model configuration (Table 2, rightmost column). Across eight model–embedding combinations, the mean ΔAUC from adding AlphaFold features was -0.001 ± 0.002, with no single comparison reaching statistical significance (paired t-test, $p > 0.05$ after Bonferroni correction). This null result does not imply that structural features are uninformative for GPCR coupling prediction in general; rather, it indicates that for this specific binary classification task under the 38 structural descriptors examined, ESM-2 650M embeddings capture information that overlaps substantially with—and renders redundant—these particular explicit structural features.

This finding aligns with the observation that PLM attention maps recover residue-residue contact patterns [42] and that ESM-2 embeddings can predict secondary structure and solvent accessibility with high accuracy [13]. From a practical standpoint, this supports the use of sequence-only features for GPCR coupling prediction at the current level of predictive performance, eliminating the computational burden of structure prediction for large-scale screening applications. Whether explicit structural features confer additional benefit for finer-grained tasks—such as subtype-level prediction or ligand-dependent coupling bias—remains an open question.

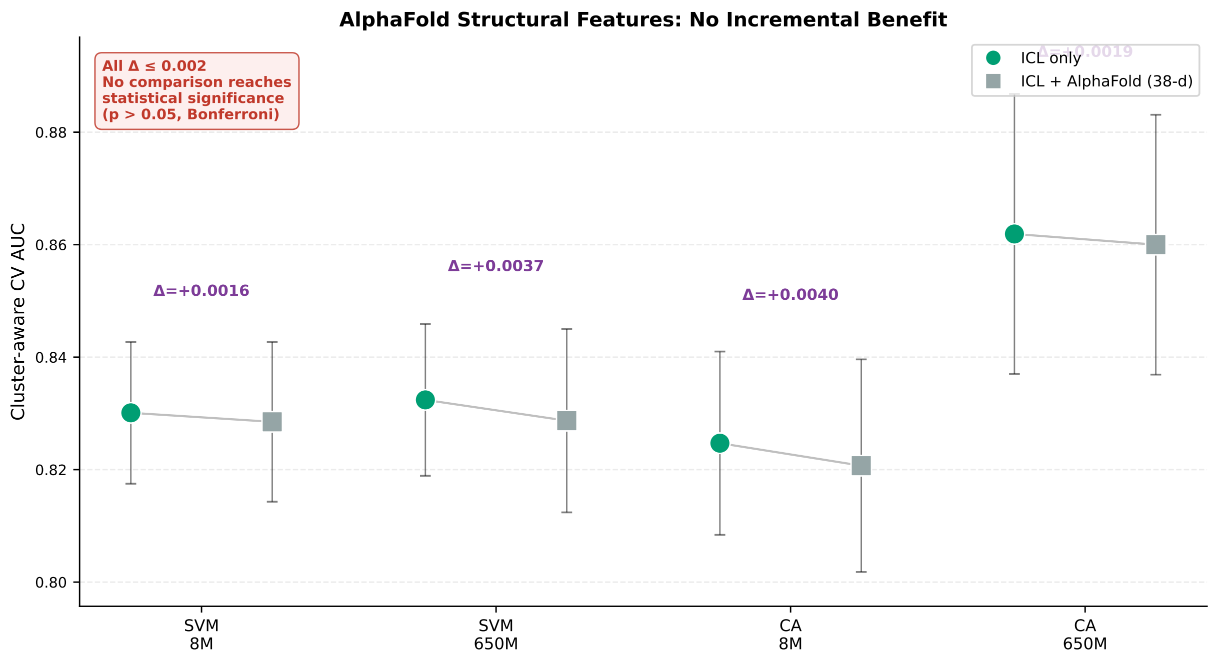

Figure 5: AlphaFold structural feature ablation. Across eight model–embedding configurations, adding 38 AlphaFold descriptors (TM distances, ICL geometry, dihedral angles, SASA, DSSP ratios, PAE features; see Supplementary Table S1) yields ΔAUC ≤ 0.002 with no individual comparison reaching statistical significance (paired t-test, $p > 0.05$ after Bonferroni correction across 8 comparisons). Results shown for cluster-aware 5-fold CV. Error bars: ± 1 std across 5 folds.

3.5 Explicit G protein family identity is the dominant G-protein-side signal

To dissect whether the paired formulation benefits from G protein sequence information or merely from explicit family identity, we conducted a controlled ablation comparing four conditions: (a) no G protein information (multi-task CA), (b) 4-dimensional one-hot family identity vector with logistic regression, (c) 4-dimensional one-hot family identity vector with MLP and SVM, and (d) full 1280-dimensional G protein sequence embedding with cross-attention and SVM.

The multi-task cross-attention variant, which removes all G protein identity from the input, achieved Cluster-CV AUC of 0.802 ± 0.019—substantially lower than the paired model (bootstrap AUC 0.847, ΔAUC ≈ 0.045), confirming that G protein family information is essential for prediction.

Replacing the G protein sequence embedding with a 4-dimensional one-hot family identity vector partially recovers performance: cross-attention achieves AUC 0.855 with one-hot versus 0.852 with the full 1280-dimensional embedding (difference not statistically significant). For SVM, however, the one-hot substitution causes a substantial degradation (0.773 versus 0.831, ΔAUC $= -0.058$), indicating that SVM relies on the G protein sequence features to a greater degree than cross-attention, which can effectively utilize a compact family indicator through its key-value attention mechanism.

Critically, a minimal family-conditioned MLP—the simplest nonlinear classifier with explicit family identity—achieves AUC 0.843 (95% CI: 0.804–0.883, repeated cluster-aware CV) with the 4-dimensional one-hot vector, within 0.004 of the best cross-attention model with full G protein embeddings (bootstrap AUC 0.847, 95% CI: 0.823–0.870). Logistic regression with the same features achieves 0.781 (0.758–

0.804), substantially lower, indicating that nonlinear feature interactions contribute non-trivially to predictive performance for this task. The AUC gap between minimal MLP and CA (ΔAUC $\approx$ 0.004) is within both models' bootstrap confidence intervals.

These results refine the interpretation of the paired formulation's advantage. The cross-attention architecture benefits primarily from explicit G protein family identity—knowledge of which family is being queried—rather than from the detailed sequence features of the G protein. A 4-dimensional one-hot vector is sufficient to distinguish among the four family-specific key/value spaces in cross-attention. The practical implication is that PLM-derived GPCR representations contribute the bulk of predictive power, explicit family conditioning provides a necessary additional signal, and architectural refinements beyond a standard MLP contribute modestly at this dataset scale. Future GPCR coupling prediction studies should include the minimal family-conditioned MLP as a reference baseline to quantify the contribution of architectural complexity.

3.6 Model calibration and prediction confidence

Beyond discriminative performance, practical deployment requires well-calibrated probability estimates for candidate prioritization. Under cluster-aware 5-fold cross-validation, the cross-attention model achieved a Brier score of 0.163, compared with 0.135 for SVM. For reference, the Brier score of a constant predictor that always outputs the dataset mean (positive class ratio $= 0.256$) is $0.256 \times 0.744 \approx 0.190$; a perfectly calibrated model achieves 0. The expected calibration error (ECE, 10 equal-width bins) was 0.141 for cross-attention versus 0.135 for SVM, indicating that both models produce similarly calibrated probability estimates on held-out test data (Figure 6). We note that the Brier scores reported here reflect the challenging cluster-aware evaluation protocol; under random split evaluation, calibration is substantially better, but such estimates overstate real-world reliability (the Random-CV vs. Cluster-CV gap is $\sim$0.11 AUC; Supplementary Table S2).

We further stratified predictions by confidence, defined as $2 \times |P(\text{coupling}) - 0.5|$. This linear metric ranges from 0 (complete uncertainty) to 1 (complete certainty): a confidence of 0.9 corresponds to a predicted probability of 0.95 or 0.05. At confidence threshold $\geq$ 0.9, covering 79.4% of predictions, the cross-attention model achieved 86.5% accuracy. At the most stringent threshold ($\geq$ 0.98, 63.8% coverage), accuracy reached 89.0%. While these values demonstrate that higher-confidence predictions are indeed more reliable, the model does not achieve near-perfect accuracy at any confidence level under cluster-aware evaluation — a realistic reflection of the task's difficulty when homology-based leakage is properly controlled.

Per-family calibration varies substantially (Figure 6C). The G12/13 family shows the lowest Brier score (0.103), but this reflects the extreme class imbalance (93.5% negative; Table 1) rather than superior calibration: a constant predictor that always outputs zero achieves Brier $= 0.065$ for this family. Across all four families, both models' Brier scores approach or exceed the family-level constant-predictor baseline, confirming that calibration remains a challenge under cluster-aware evaluation.

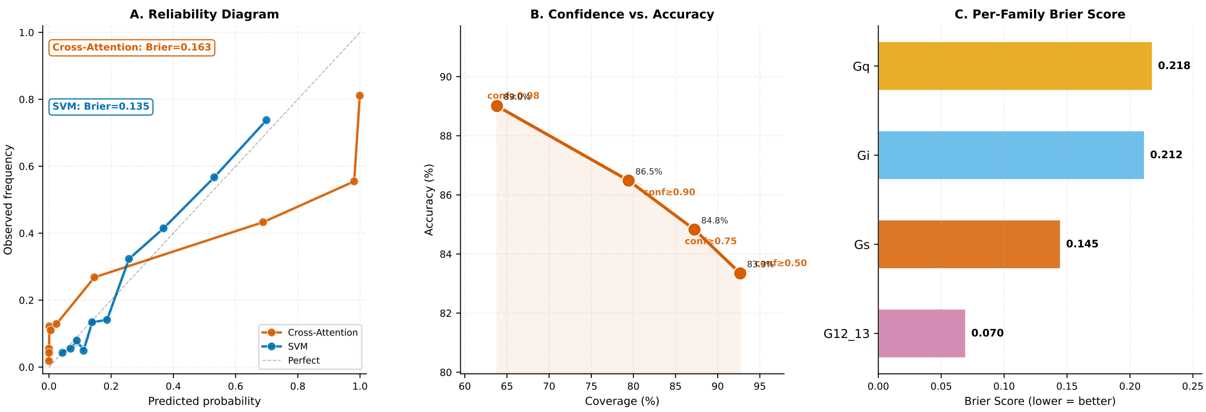

Figure 6: Model calibration and reliability analysis under cluster-aware 5-fold cross-validation. (A) Reliability diagrams comparing cross-attention (Brier = 0.163) and SVM (Brier = 0.135); constant-predictor baseline Brier = 0.190. (B) Confidence-stratified accuracy: higher-confidence predictions are more accurate, though near-perfect accuracy is not achieved. (C) Per-family Brier scores.

3.7 Annotation-quality sensitivity analysis

To assess the sensitivity of reported performance to negative label quality, we stratified the held-out CV predictions by GPCR annotation status (Table 4). Tier 1 includes all 1,639 predicted pairs. Tier 2 excludes 603 pairs from 165 zero-positive GPCRs—receptors with no experimentally validated positive coupling to any of the four families—whose negative labels may reflect lack of experimental testing rather than genuine absence of coupling.

All three models maintain substantial discriminative performance under Tier 2 (Table 4): CA achieves AUC 0.811 (vs. 0.847 in Tier 1, ΔAUC -0.035), SVM achieves 0.805 (vs. 0.831, ΔAUC -0.026), and MLP achieves 0.813 (vs. 0.838, ΔAUC -0.025). The modest degradation (ΔAUC 0.025–0.035 across models) indicates that the models capture selective coupling patterns beyond simple positive-vs.-unannotated discrimination. However, we note that the label-audit analysis in the Supplementary Materials shows that when a simplified SVM with family one-hot features (rather than full G protein embeddings) is evaluated under identical tiering, the Tier 2 AUC drops from 0.77 to 0.37—a much larger degradation that reveals the interaction between feature richness and label quality. The full models' relative robustness likely reflects the information content of high-dimensional PLM representations, which may encode coupling-relevant signals that partially compensate for noisy negative labels.

Per-family Tier 2 analysis (Supplementary Table S9) shows that performance degradation is not uniform across families. G12/13 shows the largest relative drop (CA PRAUC 0.23), consistent with its limited positive sample size (26 pairs). Gq and Gi maintain PRAUC above 0.72 under Tier 2 for all three models. We recommend that future GPCR coupling benchmarks adopt label-quality tiering as a standard reporting practice, with explicit annotation of negative evidence status (tested-negative vs. untested) in dataset releases.

3.8 Per-family performance and G12/13 practical limitations

Aggregate AUC masks substantial per-family variation (Table 5). The cross-attention model achieves per-family PRAUC of 0.655 (Gq), 0.776 (Gi), 0.629 (Gs), and 0.230 (G12/13). For the G12/13 family—the least-represented with only 26 positive pairs (6.5% positive ratio)—performance metrics reveal practical limitations that AUC alone obscures. While the per-family AUC of 0.705 appears moderate, the

Table 4: Annotation-quality tiered evaluation. Tier 1: full dataset. Tier 2: zero-positive GPCRs excluded. All metrics from cluster-aware 5-fold CV held-out predictions.

Model	Tier	N	Pos. ratio	AUC	PRAUC	Brier
CA 650M+ICL	Tier 1	1,639	0.254	0.847	0.681	0.163
	Tier 2	1,036	0.403	0.811	0.766	0.228
SVM 650M+ICL	Tier 1	1,639	0.254	0.831	0.643	0.135
	Tier 2	1,036	0.403	0.805	0.733	0.217
MLP 650M+ICL	Tier 1	1,639	0.254	0.838	0.627	0.199
	Tier 2	1,036	0.403	0.813	0.725	0.224

Tier 2 removes 165 zero-positive GPCRs (603 pairs). The minimal family-conditioned MLP achieves Tier 1 AUC 0.843 (95% CI: 0.804–0.883) from repeated cluster-aware CV; its Tier 2 performance is pending evaluation. Per-family Tier 2 breakdowns are in Supplementary Table S9.

PRAUC of 0.23 indicates poor precision-recall trade-off. Recall at 50% precision is 0.038 for CA (0.077 for SVM), meaning that fewer than 4% of true G12/13-positive pairs are recovered at a precision level suitable for experimental screening. Precision at the top-10 ranked predictions is 0.40 for CA (0.20 for SVM), and the mean predicted probability for true positive G12/13 pairs is 0.21 for CA (0.19 for SVM), indicating systematic under-confidence for this family.

These results warrant specific guidance for practical use. For Gq, Gi, and Gs families, the models provide reasonably calibrated probability estimates that can inform experimental prioritization (per-family Brier scores: 0.218, 0.212, 0.145 for CA). For G12/13, probability output should be treated strictly as exploratory ranking only and should not be used as standalone experimental prioritization scores. The low recall at usable precision reflects both the small positive sample (26 pairs) and the biological challenge: G12/13-coupled receptors share fewer discriminative sequence features with each other than receptors coupling to the other three families, consistent with the evolutionary divergence of the G12/13 subfamily [7]. Future work targeting G12/13 should consider data augmentation strategies, family-specific calibration layers, or incorporation of G12/13-specific structural interaction features.

3.9 LOGPSO remains an open community challenge

Cross-family generalization (LOGPSO) is the most stringent evaluation: models trained on three G protein families must predict coupling to the held-out fourth family. Mean LOGPSO AUC across all methods clustered tightly at ∼0.60 (SVM: 0.606, cross-attention: ∼0.60, multi-task: 0.591; Supplementary Table S3), only marginally above the random baseline of 0.50. Per-family performance varied considerably: Gs was consistently the hardest to generalize to (mean AUC 0.52), while Gi was the easiest (mean AUC 0.67), reflecting the evolutionary distances among Gα families [7].

The uniformity of LOGPSO performance across architectures with fundamentally different inductive biases—linear kernel methods, tree ensembles, deep feed-forward networks, and cross-attention—indicates that current approaches do not learn transferable coupling determinants that generalize across G protein families. The models predominantly capture family-specific patterns rather than universal interface features. This represents a fundamental limitation: the framework performs well when G protein family identity is available at inference time (paired formulation) but cannot reliably predict coupling to an unseen G protein family. We propose that LOGPSO should be adopted as a community benchmark for GPCR coupling prediction, analogous to the role of CASP in structure prediction [43], to systematically track progress on this challenging cross-family generalization problem.

Table 5: Per-family performance metrics under cluster-aware 5-fold CV. P@10: precision at top-10 ranked predictions. R@P=0.5: recall at 50% precision. MeanProb+: mean predicted probability for true positive pairs.

Family	Model	AUC	PRAUC	Brier	P@10	R@P=0.5	MeanProb+
3*Gq (N=448, 34%+)	CA	0.793	0.655	0.218	0.90	0.842	0.676
	SVM	0.771	0.622	0.180	0.90	0.829	0.446
	MLP	0.759	0.590	0.243	0.70	0.789	0.730
3*Gi (N=397, 40%+)	CA	0.846	0.776	0.212	1.00	0.975	0.787
	SVM	0.807	0.723	0.177	0.90	0.962	0.510
	MLP	0.829	0.735	0.208	0.90	0.969	0.830
3*Gs (N=397, 20%+)	CA	0.811	0.629	0.145	0.90	0.650	0.579
	SVM	0.817	0.616	0.117	0.90	0.675	0.391
	MLP	0.837	0.561	0.149	0.70	0.688	0.658
3*G12/13 (N=397, 7%+)	CA	0.705	**0.230**	0.070	0.40	**0.038**	0.214
	SVM	0.684	0.232	0.060	0.20	0.077	0.192
	MLP	0.696	0.222	0.074	0.40	0.231	0.309

Cluster-aware 5-fold CV held-out predictions. P@10 = precision among the 10 highest-ranked pairs per family. R@P=0.5 = recall at 50% precision threshold. G12/13 values in bold highlight practical screening limitations.

3.10 Feature importance implicates ICL3 as primary determinant

Integrated gradient attribution on the 650M cross-attention model revealed a strong asymmetry: GPCR-side features exhibited 5- to 7-fold higher aggregate attribution than G-protein-side features (Supplementary Figure S1). Within GPCR features, ICL3 physicochemical statistics showed the highest per-group sensitivity (mean absolute attribution 0.18), followed by ICL2 statistics (0.14) and global GPCR embedding (0.11). This hierarchy is biologically plausible: ICL3 is the primary structural determinant of G protein coupling selectivity, with its length, charge distribution, and conformational dynamics directly influencing which Gα subtype is engaged [26, 27].

4 Case study: Coupling prioritization for under-characterized GPCRs

To illustrate the model's utility for hypothesis generation, we applied the trained cross-attention 650M model to predict coupling probabilities for GPCRs with limited experimental characterization. From the dataset, we identified 42 GPCRs that have either no experimentally validated G protein coupling in GPCRdb/IUPHAR or incomplete coupling annotation (annotated against $\leq$1 G protein family). These predictions are coupling prioritization hypotheses for experimental follow-up, not validated couplings, and should be interpreted within the framework of homology-guided inference.

The model assigns differentiated coupling probability profiles across families for several under-characterized GPCRs (Supplementary Table S8). For example, GPRC5B (Q8TDV5), a Class C receptor with emerging evidence for non-canonical signaling, receives a high Gs probability (0.999) but low probabilities for the other three families—a selectivity pattern consistent with the limited functional data available for this receptor. GPRC5C (Q14832), a closely related Class C receptor, shows a distinct predicted profile (high Gi, moderate Gq), demonstrating that the model discriminates among receptors within the same subfamily. Across the 42 under-characterized GPCRs, the model assigns at least one family probability >0.7 for 31 receptors (74%), providing a ranked list for experimental screening prioritization.

Three interpretive caveats apply. First, many of these GPCRs share substantial sequence homology with well-characterized receptors. For example, GPRC5B shares the Class C 7TM fold with the calcium-sensing receptor and metabotropic glutamate receptors, which couple primarily to Gq and Gi. The model's predictions for such receptors reflect learned sequence homology patterns with their characterized paralogs—a form of *homology-guided inference* rather than ab initio prediction. For this reason, predictions for receptors with close, well-characterized homologs should be interpreted as paralog-transfer hypotheses, and experimental follow-up should prioritize receptors with *low* sequence identity to training-set GPCRs sharing the predicted coupling family, as these candidates better demonstrate the model's value beyond simple sequence-similarity transfer. Second, given the model's moderate calibration under cluster-aware evaluation and the poor precision-recall performance for G12/13 (detailed in the Annotation-quality sensitivity analysis and Per-family performance sections), probabilities should be treated as relative ranking scores within each family rather than absolute confidence estimates. Third, temporal validation—training on data available before a cutoff date and evaluating on couplings confirmed afterward—would provide a more rigorous assessment of prospective predictive power but was not attempted here because the dataset lacks publication timestamps for individual coupling annotations.

5 Discussion

5.1 Methodological contributions

This work makes five contributions, ordered by their significance for the GPCR coupling prediction community.

1. Leakage-controlled benchmark. The cluster-aware evaluation protocol prevents the $\sim$0.11 AUC inflation observed under random splitting (Supplementary Table S2), a methodological safeguard that prior GPCR coupling prediction studies have not uniformly adopted. The standardized Cluster-CV and LOGPSO protocols, implemented on a curated dataset of 1,647 pairs across 387 sequence-homology clusters, provide a reproducible foundation for tracking progress.

2. Annotation-quality audit. The label-quality tiered evaluation (Section 3.7) reveals that performance estimates are sensitive to the inclusion of zero-positive GPCRs—receptors with no experimentally validated coupling to any family. While the full models maintain substantial discriminative performance after removing these GPCRs (ΔAUC 0.025–0.035), a simplified SVM with family one-hot features drops from AUC 0.77 to 0.37 (Supplementary Materials), demonstrating that feature richness interacts with label quality. We recommend that future dataset releases annotate negative labels with explicit evidence status (tested-negative vs. untested) and that benchmarks adopt label-quality tiering as a standard reporting practice.

3. Family-conditioned PLM baseline. The empirical demonstration that explicit G protein family identity is the dominant signal from the G protein side is practically consequential. Replacing the 1280-dimensional G protein sequence embedding with a 4-dimensional one-hot family vector leaves cross-attention AUC essentially unchanged (0.855 vs. 0.852; Supplementary Table S6). A minimal family-conditioned MLP—the simplest architecture with explicit family conditioning—achieves AUC 0.843 (95% CI: 0.804–0.883), within 0.004 of the best cross-attention model (bootstrap AUC 0.847, 95% CI: 0.823–0.870). Future GPCR coupling prediction studies should include this minimal baseline to quantify the marginal contribution of architectural complexity beyond strong PLM representations and explicit family conditioning.

4. Architecture comparison and feature ablation. Systematic comparison across five model classes, two embedding scales, and four feature configurations shows that cross-attention provides a small but statistically significant improvement over SVM (cluster-bootstrap ΔAUC 0.016, 95% CI: 0.004–0.028, $p = 0.006$) while being statistically comparable to MLP (ΔAUC 0.009, 95% CI: -0.005 to 0.024, $p = 0.133$). The empirical confirmation of a dimension alignment requirement for local-global PLM feature fusion (Section 3.3) carries practical implications for multi-scale feature engineering: control experiments (Supplementary Materials, Dimension Alignment Controls) confirm that PCA projection to a lower dimension degrades performance (AUC -0.040), while learned linear projection partially recovers it (AUC -0.006 vs. matched), indicating that dimension alignment can be addressed through projection layers when native matching is infeasible. The systematic ablation of 38 AlphaFold structural descriptors demonstrates that ESM-2 650M embeddings capture information that overlaps substantially with these particular intracellular-interface proxy features for this specific binary family-level task; finer-grained prediction problems may benefit from explicit structural features.

5. Open-source tool. The `gpcr-coupling` package with pre-trained models, configurable hyperparameters, and a tutorial workflow enables reproduction and extension with minimal computational overhead.

5.2 Comparison with related work

A direct numerical comparison with prior work requires caution due to differences in evaluation protocol. [20] reported an AUC of 0.87 using AlphaFold3-derived interface features and a single-protein classifier. However, their evaluation used random split cross-validation, whereas our primary results employ cluster-aware splitting—which prevents GPCRs sharing sequence identity above the clustering threshold from appearing in both training and test sets. On our data, the gap between Random-CV and Cluster-CV is ∼0.11 AUC for the same model (Supplementary Table S2). Because their evaluation protocol, dataset composition, and feature set all differ from ours, we refrain from applying post-hoc correction factors and instead report both Random-CV (0.970) and Cluster-CV (0.862) results to facilitate future direct comparisons under standardized protocols. Our model uses only sequence-derived ESM-2 embeddings, while theirs requires per-protein AlphaFold3 structure prediction, a difference with practical implications for large-scale screening throughput.

The architecture comparison across five model classes provides a reference point for future GPCR coupling prediction studies. The pooled-feature cross-attention architecture used here differs fundamentally from residue-level cross-attention developed for protein–protein interaction prediction [18]: our cross-attention operates on single summary vectors (mean-pooled embeddings projected to a shared space) rather than per-residue representations. The modest but statistically significant improvement of cross-attention over SVM (cluster-bootstrap 95% CI for ΔAUC: 0.004–0.028, $p = 0.006$) indicates that learned nonlinear feature fusion provides a small but reliable benefit when high-dimensional (1280-d) representations are available. Cross-attention is statistically comparable to MLP (ΔAUC 0.009, 95% CI: -0.005 to 0.024, $p = 0.133$). Sequence-based PPI prediction methods have evolved from coevolutionary analysis [44] to deep learning architectures [17, 18]. Following the precedent of large-scale protein–protein interaction benchmarks [45], we encourage reporting under both Cluster-CV and LOGPSO protocols to enable fair comparison across studies.

5.3 Limitations and future directions

LOGPSO generalization. The ∼0.60 AUC ceiling under LOGPSO is the primary limitation. Promising directions include: (i) G protein family-specific embedding layers that learn transferable representations of family-level features, (ii) few-shot meta-learning where LOGPSO families serve as meta-test tasks during training, (iii) incorporation of evolutionary coupling features from multiple sequence alignments across species, and (iv) explicit structural docking or interface energy terms that are family-independent. We encourage the community to treat LOGPSO as a benchmark task and report per-family decomposition to enable tracking progress on the hardest sub-problems.

Dataset coverage. Our dataset of 431 GPCRs represents $\sim$53% of the non-olfactory human GPCRome. Furthermore, the four G protein families are unevenly represented: G12/13 contributes only 53 GPCRs (12.3% of the dataset, 105 pairs), compared with 308 GPCRs for Gi/o. Performance estimates for G12/13, especially under LOGPSO, should be interpreted with caution due to limited sample size. Future expansion to additional receptors—particularly Class B (secretin), Class C (glutamate), and Frizzled/Taste2 families—as experimental coupling data become available will increase both coverage and biological diversity.

Cluster granularity in cross-validation. CD-HIT clustering at 40% sequence identity produced 387 clusters from 431 GPCRs, meaning approximately 90% of clusters are singletons. While this conservative threshold minimizes homology leakage, the high singleton proportion implies that Cluster-CV performance may approach Random-CV more closely than the protocol intends—the primary effect of cluster-aware splitting is to isolate the $\sim$10% of GPCRs that do share $\geq$40% identity with another receptor in the dataset. A sensitivity analysis using multiple identity thresholds (e.g., 40%, 60%, 90%) would characterize how sequence identity cutoffs affect generalization estimates; for example, a 90% threshold, which would group only near-identical sequences, could reveal whether performance degradation is driven primarily by close homologs or by more subtle sequence similarity. We provide cluster assignments at the 40% threshold in the companion repository and encourage re-evaluation at alternative thresholds in future benchmarking studies.

Orthosteric and allosteric ligands. The current model predicts basal coupling preference, but ligand-dependent coupling bias (functional selectivity) is pharmacologically important. Extending the paired formulation to include ligand features—either chemical fingerprints or learned molecular representations—could enable prediction of ligand-specific coupling profiles.

Binary labeling simplification and negative label quality. Our binary classification framework treats all coupling annotations—whether derived from weak, borderline BRET signals or robust, multi-assay consensus—as equivalent positive labels. Coupling efficiency is inherently graded, and the binary simplification may conflate biologically distinct interaction strengths. Equally important, negative labels are heterogeneous: of the 1,226 negative pairs, approximately 584 (48%) originate from GPCRs that have zero positive annotations across all four families (165 "zero-positive" GPCRs). For these GPCRs, negative labels may reflect genuine absence of coupling, lack of experimental testing against specific families, or coupling to G protein families outside the four-family scope. A label audit (Supplementary Materials) reveals that when these zero-positive GPCRs are excluded, SVM performance with the minimal family-conditioned features drops from AUC 0.77 to 0.37, indicating that a substantial fraction of discriminative performance derives from distinguishing annotated-positive GPCRs from completely unannotated GPCRs rather than from resolving selective coupling patterns. We therefore report results under three label-quality tiers in the Supplementary Materials and encourage future dataset curation efforts to annotate negatives with explicit evidence status (tested-negative vs. untested). Quantitative coupling measures such as $\Delta \log(E_{\max}/EC_{50})$ from BRET assays would enable regression-based prediction and mitigate the binary labeling simplification. The current binary formulation provides a foundation that can be extended as quantitative coupling data become available at scale.

G12/13 family: per-family metrics reveal practical limitations. The G12/13 family contributes only 26 positive pairs (6.5% positive ratio; Table 1). While the cross-attention model achieves per-family AUC 0.705 for this family, the precision-recall AUC is 0.23, and recall at 50% precision is 0.038 (Table 5). Mean predicted probability for true G12/13-positive pairs is 0.21, indicating that the model systematically underestimates coupling probability for this family. The seemingly reasonable AUC reflects the model's ability to rank the 26 positive pairs above a random subset of the 371 negatives, but the probability estimates are not practically useful for screening applications. We recommend that G12/13 probability output be treated strictly as exploratory ranking and not used as standalone experimental prioritization scores. Future work targeting the G12/13 family should consider dedicated modeling strategies such as data augmentation, family-specific calibration layers, or incorporation of G12/13-specific interaction features.

5.4 Practical guidelines

From this systematic evaluation, three practical guidelines emerge for GPCR coupling prediction. The paired formulation with explicit G protein identity should be preferred over single-protein classification wherever G protein sequence information is available. When combining PLM embeddings across scales, local and global feature dimensions should be matched natively or aligned through learned projection layers. Before committing to computationally expensive structure prediction pipelines, practitioners should benchmark explicit structural features against a PLM-only baseline, since ESM-2 650M may already capture information that overlaps with standard structural descriptors for family-level coupling classification.

5.5 Tutorial: Reproducing and extending the benchmark

To facilitate adoption and extension, we provide a step-by-step tutorial in the software documentation. The complete reproduction workflow consists of four commands: (1) `gpcr-coupling extract-features` to generate ESM-2 and ICL features from input FASTA files, (2) `gpcr-coupling train` to reproduce the cross-attention model with configurable hyperparameters, (3) `gpcr-coupling predict` to generate coupling probabilities for novel (GPCR, G protein) pairs, and (4) `gpcr-coupling evaluate` to compute AUC, PRAUC, Brier score, and ECE against a labeled test set. All pre-computed features and model weights are distributed with the package, enabling reproduction in under 30 minutes on a consumer GPU.

6 Conclusion

This study establishes a leakage-controlled benchmark for family-level GPCR–G protein coupling prediction and makes five contributions: (i) a cluster-aware evaluation protocol that prevents the $\sim$0.11 AUC inflation from random splitting; (ii) an annotation-quality audit demonstrating that label quality interacts with feature richness and that future benchmarks should adopt tiered reporting; (iii) a minimal family-conditioned MLP baseline (AUC 0.843) that approaches the best model (AUC 0.847) and demonstrates that explicit family identity is the dominant G-protein-side signal; (iv) a systematic architecture comparison showing that cross-attention provides a small but statistically significant improvement over SVM (ΔAUC 0.016, $p = 0.006$) while being statistically comparable to MLP ($p = 0.133$), that dimension matching is required for local-global PLM feature fusion, and that 38 intracellular-interface proxy features from monomer AlphaFold2 structures contribute no incremental value beyond ESM-2 650M embeddings for this binary family-level task; and (v) an open-source tool with pre-trained models. Practical limitations include poor G12/13 precision-recall performance (PRAUC 0.23, recall at 50% precision < 0.04) motivating explicit per-family reporting, and the $\sim$0.60 AUC ceiling under cross-family

LOGPSO evaluation. The companion benchmark dataset, standardized protocols, and software establish a reproducible foundation for tracking progress on GPCR–G protein coupling specificity prediction.

Data availability

All data, code, and pre-trained models supporting this study are available at https://github.com/nblvguohao/gpcr-coupling-leakage-benchmark under the MIT license. The repository name reflects the methodological focus on preventing homology-induced data leakage through cluster-aware evaluation—a central lesson from this benchmark. The paired dataset (1,647 pairs, 431 GPCRs) is provided in CSV format with corresponding FASTA sequences. Pre-computed ESM-2 embeddings (8M and 650M) for all GPCRs and G proteins, ICL features, and AlphaFold structural descriptors are included to enable reproduction without GPU requirements for feature extraction. Cross-validation fold assignments and cluster membership files are provided to ensure exact reproducibility of all reported metrics.

Author contributions

G.L., Y.X., and H.L. conceived the study and designed the experimental framework. G.L. implemented the models, performed computational analyses, and developed the software tool. H.L. contributed to data curation and feature extraction. X.Z. and S.Y. contributed to methodology refinement and result interpretation. L.G. and Q.W. supervised the project. All authors contributed to manuscript preparation, reviewed the results, and approved the final version.

Acknowledgements

We thank the GPCRdb team for maintaining the curated GPCR coupling database, and the ESM development team at Meta AI for releasing pre-trained protein language models. Computational resources were provided by the High-Performance Computing Center at Anhui Agricultural University.

Funding

This work was supported by grants from the National Natural Science Foundation of China (32472007, 62301006, 62301008), the Natural Science Foundation of Anhui Province (2308085MF217, 2308085QF202), and the Anhui Province Key Laboratory of Intelligent Agricultural Technology and Equipment.

Conflict of interest

The authors declare no competing financial or non-financial interests.

References

[1] Kristen L Pierce, Richard T Premont, and Robert J Lefkowitz. Seven-transmembrane receptors. *Nature Reviews Molecular Cell Biology*, 3(9):639–650, 2002. doi: 10.1038/nrm908.

[2] Daniel M Rosenbaum, Søren G F Rasmussen, and Brian K Kobilka. The structure and function of G-protein-coupled receptors. *Nature*, 459(7245):356–363, 2009. doi: 10.1038/nature08144.

[3] Alexander S Hauser, Misty M Attwood, Mikael Rask-Andersen, Helgi B Schiöth, and David E Gloriam. Trends in GPCR drug discovery: new agents, targets and indications. *Nature Reviews Drug Discovery*, 16(12):829–842, 2017. doi: 10.1038/nrd.2017.178.

[4] Krishna Sriram and Paul A Insel. G protein-coupled receptors as targets for approved drugs: how many targets and how many drugs? *Molecular Pharmacology*, 93(4):251–258, 2018. doi: 10.1124/mol.117.111062.

[5] Xi-Ping Tang et al. Orphan GPCRs: structural biology and drug discovery. *Trends in Pharmacological Sciences*, 44(9):586–601, 2023. doi: 10.1016/j.tips.2023.06.005.

[6] Simon R Foster, Alexander S Hauser, Line Vedel, Ryan T Strachan, Xi-Ping Huang, Amadeu Gavaldà, Gáspár Pándy-Szekeres, David E Gloriam, and Hans Bräuner-Osborne. Discovery of human signaling systems: pairing peptides to G protein-coupled receptors. *Cell*, 179(4):895–908.e21, 2019. doi: 10.1016/j.cell.2019.10.010.

[7] Nina Wettschureck and Stefan Offermanns. Mammalian G proteins and their cell type specific functions. *Physiological Reviews*, 85(4):1159–1204, 2005. doi: 10.1152/physrev.00003.2005.

[8] Reid H J Olsen, Jeffrey F DiBerto, Justin G English, Alexis M Glaudin, Brian E Krumm, Samuel T Slocum, Tao Che, Ariana C Gavin, John D McCorvy, Bryan L Roth, et al. TRUPATH, an open-source biosensor platform for interrogating the GPCR transducerome. *Nature Chemical Biology*, 16(8):841–849, 2020. doi: 10.1038/s41589-020-0535-8.

[9] Steen Möller, Jaak Vilo, and Michael D R Croning. Prediction of the coupling specificity of G protein-coupled receptors to their G proteins. *Bioinformatics*, 17(Suppl 1):S174–S181, 2001. doi: 10.1093/bioinformatics/17.suppl_1.s174.

[10] Toshinori Ono and Haretsugu Hishigaki. Prediction of GPCR-G protein coupling specificity using features of sequences and biological functions. *Genomics, Proteomics and Bioinformatics*, 4(1): 26–35, 2006. doi: 10.1016/S1672-0229(06)60014-1.

[11] Nikolaos G Sgourakis, Pantelis G Bagos, Panagiotis K Papasaikas, and Stavros J Hamodrakas. A method for the prediction of GPCRs coupling specificity to G-proteins using refined profile hidden Markov models. *BMC Bioinformatics*, 6:104, 2005. doi: 10.1186/1471-2105-6-104.

[12] Y Yabuki, T Muramatsu, T Hirokawa, H Mukai, and M Suwa. GRIFFIN: a system for predicting GPCR-G-protein coupling selectivity using a support vector machine and a hidden Markov model. *Nucleic Acids Research*, 33(Web Server issue):W148–W153, 2005. doi: 10.1093/nar/gki495.

[13] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

[14] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C Lawrence Zitnick, Jerry Ma, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021. doi: 10.1073/pnas.2016239118.

[15] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, et al. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381.

[16] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30:5998–6008, 2017. doi: 10.5555/3295222.3295349.

[17] Somaye Hashemifar, Behnam Neyshabur, Aly A Khan, and Jinbo Xu. Predicting protein-protein interactions through sequence-based deep learning. *Bioinformatics*, 34(17):i802–i810, 2018. doi: 10.1093/bioinformatics/bty573.

[18] Samuel Sledzieski, Rohit Singh, Lenore Cowen, and Bonnie Berger. D-SCRIPT translates genome to phenome with sequence-based prediction of protein-protein interactions. *Nature Communications*, 12(1):5300, 2021. doi: 10.1038/s41467-021-25554-4.

[19] Muhao Chen, Chelsea J-T Ju, Guangyu Zhou, Xuelu Chen, Tianran Zhang, Kai-Wei Chang, Carlo Zaniolo, and Wei Wang. Multifaceted protein-protein interaction prediction based on Siamese residual RCNN. *Bioinformatics*, 35(14):i305–i314, 2019. doi: 10.1093/bioinformatics/btz328.

[20] P Miglionico et al. Predicting and engineering GPCR-G protein coupling specificity with AlphaFold3. *Bioinformatics (ISMB/ECCB 2025)*, 2025. In press.

[21] Albert J Kooistra, Stefan Mordalski, Gáspár Pándy-Szekeres, Mauricio Esguerra, Azat Mamyrbekov, Christian Munk, György M Keserű, and David E Gloriam. GPCRdb in 2021: integrating GPCR sequence, structure and function. *Nucleic Acids Research*, 49(D1):D335–D343, 2021. doi: 10.1093/nar/gkaa1080.

[22] Gáspár Pándy-Szekeres, Jeannette Caroli, Azat Mamyrbekov, György M Keserű, Albert J Kooistra, and David E Gloriam. GPCRdb in 2023: state-specific structure models using AlphaFold2 and new resources. *Nucleic Acids Research*, 51(D1):D395–D402, 2023. doi: 10.1093/nar/gkac1013.

[23] The UniProt Consortium. UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D523–D531, 2023. doi: 10.1093/nar/gkac1052.

[24] Simon D Harding, Jane F Armstrong, Elena Faccenda, Christopher Southan, Stephen P H Alexander, Anthony P Davenport, Adam J Pawson, Michael Spedding, and Jamie A Davies. The IUPHAR/BPS Guide to Pharmacology in 2024. *Nucleic Acids Research*, 52(D1):D1438–D1449, 2024. doi: 10.1093/nar/gkad944.

[25] Nicklas S Detlefsen, Søren Hauberg, and Wouter Boomsma. Learning meaningful representations of protein sequences. *Nature Communications*, 13:1914, 2022. doi: 10.1038/s41467-022-29443-w.

[26] Tilman Flock, Alexander S Hauser, Nadia Lund, David E Gloriam, Santhanam Balaji, and M Madan Babu. Selectivity determinants of GPCR-G-protein binding. *Nature*, 545(7654):317–322, 2017. doi: 10.1038/nature22070.

[27] Søren G F Rasmussen, Brian T DeVree, Yaozhong Zou, Andrew C Kruse, Ka Young Chung, Tong Sun Kobilka, Foon Sun Thian, Pil Seok Chae, Els Pardon, Diane Calinski, et al. Crystal structure of the $\beta2$ adrenergic receptor–Gs protein complex. *Nature*, 477(7366):549–555, 2011. doi: 10.1038/nature10361.

[28] Jeppe Hallgren, Konstantinos D. Tsirigos, Mads Damgaard Pedersen, Jose Juan Almagro Armenteros, Paolo Marcatili, Henrik Nielsen, Anders Krogh, and Ole Winther. DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks. *bioRxiv*, 2022. doi: 10.1101/2022.04.08.487609. Preprint.

[29] Shuichi Kawashima, Piotr Pokarowski, Maria Pokarowska, Andrzej Kolinski, Toshiaki Katayama, and Minoru Kanehisa. AAindex: amino acid index database, progress report 2008. *Nucleic Acids Research*, 36(Database issue):D202–D205, 2008. doi: 10.1093/nar/gkm998.

[30] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.

[31] Mihaly Varadi, Damian Bertoni, Paul Magar, Uddipan Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Svetlana Tsilova, Milot Mirdita, et al. AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*, 52(D1):D368–D375, 2024. doi: 10.1093/nar/gkad1011.

[32] Wolfgang Kabsch and Christian Sander. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22(12):2577–2637, 1983. doi: 10.1002/bip.360221211.

[33] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[34] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*, 2016.

[35] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*, 2019.

[36] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

[37] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

[38] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: an imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32:8026–8037, 2019.

[39] Jesse Davis and Mark Goadrich. The relationship between Precision-Recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pages 233–240, 2006. doi: 10.1145/1143844.1143874.

[40] Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015. doi: 10.1371/journal.pone.0118432.

[41] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 3319–3328, 2017.

[42] Roshan Rao, Joshua Meier, Tom Sercu, Sergey Ovchinnikov, and Alexander Rives. Transformer protein language models are unsupervised structure learners. *Proceedings of the National Academy of Sciences*, 118(49):e2021160118, 2021. doi: 10.1073/pnas.2021160118.

[43] Andriy Kryshtafovych, Torsten Schwede, Maya Topf, Krzysztof Fidelis, and John Moult. Critical assessment of methods of protein structure prediction (CASP)—round XV. *Proteins: Structure, Function, and Bioinformatics*, 91(12):1539–1549, 2023. doi: 10.1002/prot.26617.

[44] Qian Cong, Ivan Anishchenko, Sergey Ovchinnikov, and David Baker. Protein interaction networks revealed by proteome coevolution. *Science*, 365(6449):185–189, 2019. doi: 10.1126/science.aaw6718.

[45] R Reim et al. Large-scale benchmark of sequence-based protein-protein interaction prediction. *Nature Methods*, 22(3):485–494, 2025. doi: 10.1038/s41592-024-02561-5.